# TRANSFORMING SPOTIFY DATA: BTS'S ROAD TO 1B STREAMS

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## PROJECT DESCRIPTION

- Analyzes the Spotify streaming performance of the K-pop group BTS, inspired by the annual Spotify Wrapped.
- Focuses on the group's current popularity and identifies their most-streamed songs on the platform.
- The scope is limited to tracks released under the official "BTS" artist profile, excluding solo projects or external collaborations.

## OBJECTIVE

- To identify which of BTS's top songs have surpassed 1 billion streams to join the "Billions Club."
- To forecast which songs are closest to this milestone and project an estimated timeline based on current daily streaming data.

## METHODOLOGY

- Data Extraction: pulled artist/track metadata from the Spotify API via the spotipy library and stream counts from a Kworb.net CSV file.
- Data Transformation: cleaned and merged the two datasets, then engineered new features like avg_daily_streams and a momentum_score to analyze performance trends.
- Data Loading: loaded the final, clean dataframes into a local SQLite relational database for structured querying and storage.

**Figure 1.** Final DataFrame after ETL process

| | artist_name | track_name | release_date | total_streams | daily_streams | avg_daily_streams | momentum_score | track_popularity | followers | genre |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | BTS | Dynamite | 2020-11-20 | 2130562214 | 475419 | 1192924.0 | 0.40 | 79 | 80988351 | k-pop |
| 1 | BTS | My Universe | 2021-10-15 | 1512971721 | 365387 | 1038416.0 | 0.35 | 76 | 80988351 | k-pop |
| 2 | BTS | Butter | 2022-06-10 | 1391090445 | 248640 | 1141173.0 | 0.22 | 74 | 80988351 | k-pop |
| 3 | BTS | Boy With Luv (feat. Halsey) | 2019-04-12 | 1309755727 | 292279 | 551708.0 | 0.53 | 76 | 80988351 | k-pop |

## KEY FINDINGS

- BTS has over 80 million Spotify followers with a platform popularity score of 85.
- Six of their top 10 most popular tracks have now surpassed 1 billion streams, officially joining the "Billions Club."
- "Life Goes On" has the highest momentum score (1.01), indicating its current daily stream rate is slightly higher than its historical average.
- A strong positive correlation was found between Spotify's internal track_popularity score and the actual daily_streams from Kworb.net.

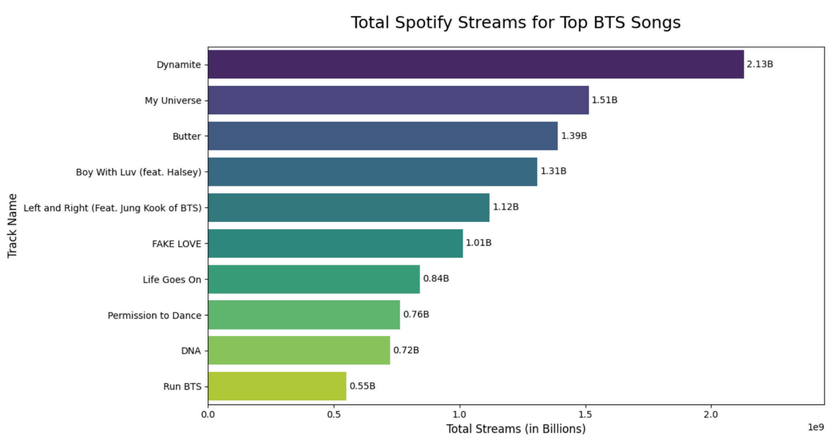


**Figure 2.** Total streams for BTS's top 10 songs

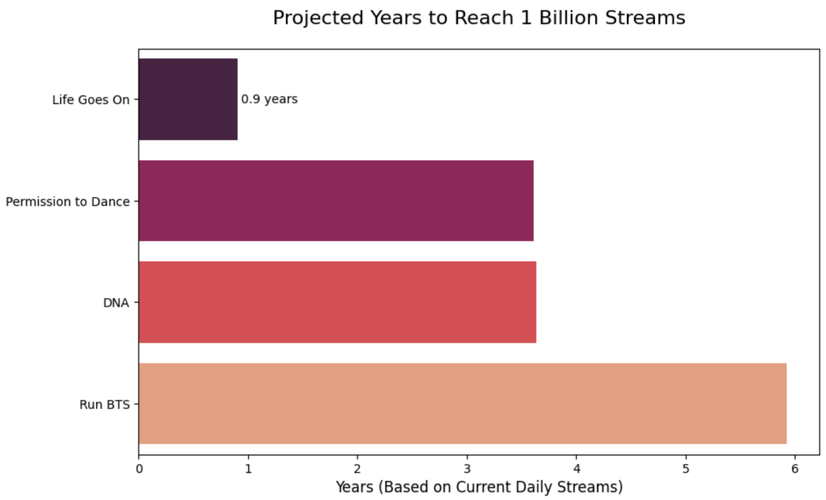


**Figure 3.** Projected time for remaining songs to hit 1B streams

## FUTURE WORK

- Time-Series Analysis: implement daily data fetching to analyze how streaming numbers fluctuate over time in response to events like new releases or social media trends.
- Audio Feature Analysis: extract audio features (e.g., danceability, acousticness) to identify common sonic characteristics among BTS's most popular songs.
- Predictive Modeling: train a classification model (e.g., Logistic Regression) to predict a new song's potential to join the "Billions Club" based on its audio features.

## REFERENCES

- Google. (2025). Gemini (Large language model).
- Kworb.net. (n.d.). BTS - Spotify Top Songs. Retrieved October 11, 2025, from https://kworb.net/spotify/artist/3NrfpeOtUJi4K4DXYWgMUX_songs.html
- Spotify. (n.d.). BTS. Retrieved October 11, 2025, from https://open.spotify.com/artist/3NrfpeOtUJi4K4DXYWgMUX